

PULSE HEALTH: OFFLINE-FIRST ARCHITECTURE

CRDT Reconciliation and Zero-Knowledge Biometric Encryption

1. PROBLEM: Ensuring healthcare tracking works reliably in low-connectivity conditions.
2. CLIENT DATABASE: Embedded SQLite running local-first on mobile and desktop devices.
3. SYNC ENGINE: State-based CRDTs resolving concurrent edits with deterministic merge logic.
4. PRIVACY: Client-side AES-256-GCM encryption with hardware keychain key storage.